BYoung Physical Therapy

Patient: John Jiles • **Date:** November 09, 2026
Surgical Side: Left • **Surgery:** 04/21/2026 (6.6 mo post-op) • **Triple Hop:** Surg 22'0" / NS 22'7" / LSI 97.4%

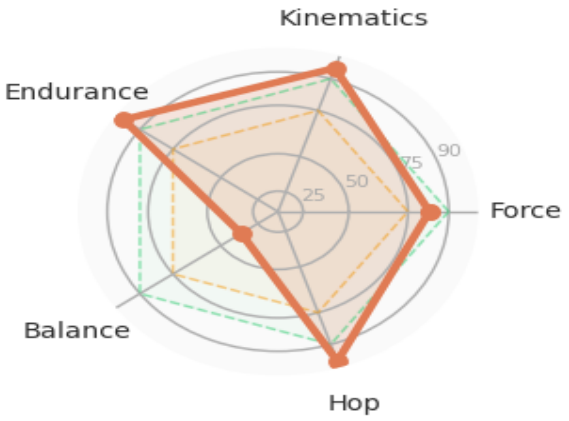
Return to Sport Assessment

44

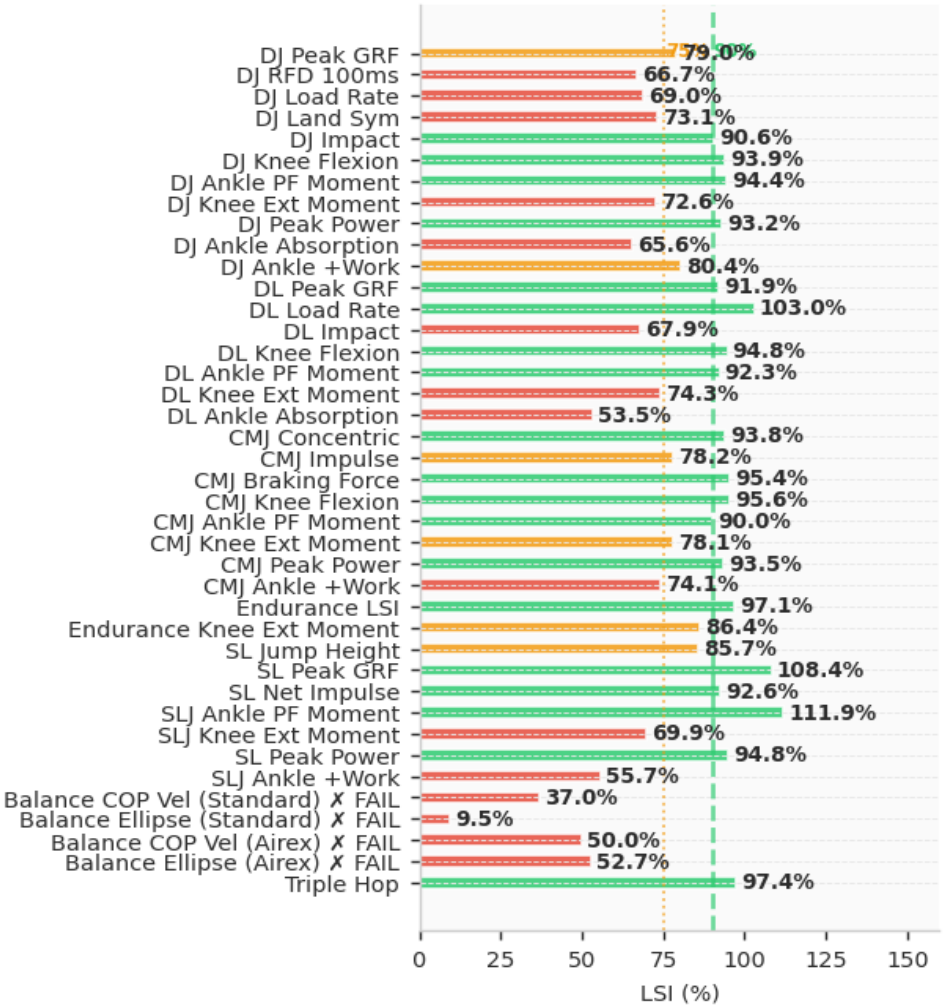
RTS Score
Not Ready

4 scored red flags · 6.6 mo post-op

Return to Sport Profile



Limb Symmetry Index — All Tests



▲ STRENGTHS

- Fatigue drift (100/100)
- DL Load rate sym. (100/100)
- Endurance symmetry (91/100)
- SL peak power (89/100)

▼ DETRACTORS

- Balance symmetry (30/100)
- Balance symmetry (Airex) (37/100)
- Landing symmetry (48/100)
- Rate of force dev. (48/100)
- DJ Peak GRF sym. (59/100)

Performance 69/100 → ×0.77 red-flag penalty (4 scored) → ×0.83 time modifier → = 44/100

CLINICAL FLAGS

- Balance (Airex): FAILED stance hold — surgical (foot moved), non-surgical (foot moved)
- Balance (Standard): FAILED stance hold — surgical (foot moved + touchdown), non-surgical (touchdown)
- Balance Ellipse (Standard) ✗ FAIL: LSI 9.5% — significant asymmetry
- Balance COP Vel (Standard) ✗ FAIL: LSI 37.0% — significant asymmetry
- DL Knee Valgus: Δ +6.6° — exceeds 6° threshold

• +15 more — see individual test pages.

Body weight 212 lb from the plates vs 190 lb in the Visual3D session — plate value used; fix the session weight (it scales joint moments).

● ≥90% Symmetric ● ≥75% Progressing ● <75% Deficit LSI = Surgical ÷ Non-Surgical × 100 | Δ = Surgical – Non-Surgical (absolute difference; used for small-angle kinematics)

DROP JUMP

ABOUT THIS TEST

The Drop Jump is a reactive bilateral landing task performed from a standard box height. The athlete drops off the box, absorbs the impact with both limbs simultaneously, then immediately jumps for maximum height. It tests the athlete's ability to rapidly store and release elastic energy (reactive strength) and to absorb high-impact loads symmetrically. Key concerns after ACL reconstruction include stiff-knee landing (reduced flexion, elevated impact transient), medial knee collapse (valgus), and side-to-side force asymmetries that indicate ongoing limb offloading. The Reactive Strength Index (RSI = jump height ÷ contact time) summarises the explosive reactive capacity of the neuromuscular system.

Metric	Surgical	Non-Surg	LSI / Δ
Peak GRF (N)	1311.8	1659.9	79.0%
Impulse (N·s)	464.1	552.4	84.0%
RFD 100ms (N/s)	19666.6	29483.9	66.7%
Loading Rate 50ms (N/s)	17307.7	25071.5	69.0%
Landing Sym 200ms (%) *(bilateral)*	—	—	73.1%
Impact Transient (N)	815.7	900.2	90.6%
Time-to-Peak (ms)	167.4	70.7	—
Jump Height (in)	23.9	—	—
Contact Time (s)	0.482	—	—
RSI (m/s)	1.26	—	—
Peak Knee Flexion (°)	85.9	91.4	93.9%
Peak Knee Valgus (°)	8.2	3.0	Δ +5.2°
Peak Tibial IR (°)	-6.4	-0.4	Δ -6.0°
Tibial Rot Excursion (°)	6.4	2.7	Δ +3.7°
Peak Hip Flexion (°)	79.6	82.9	Δ -3.2°
Peak Hip Adduction (°)	-2.8	-0.4	Δ -2.4°
Peak Hip IR (°)	11.7	4.4	Δ +7.3°
Peak Ankle Dorsiflexion (°)	26.4	29.5	Δ -3.1°
Peak Plantarflexor Moment (Nm)	174.8	185.2	94.4%
Peak Knee Ext Moment (Nm)	143.6	197.8	72.6%
AP COP Excursion (mm)	116.9	134.0	—
Trials analysed	3	—	—
Peak Power (W/kg)	71.8	—	—
Limb Peak Power (W)	3333.0	3576.4	93.2%
Limb Push-off Work (J)	536.2	580.2	92.4%
Implied Drop Height (cm)	29.0	—	—
Ankle Absorption (J)	50.1	76.4	65.6%
Knee Absorption (J)	62.3	119.9	51.9%
Hip Absorption (J)	98.4	118.6	83.0%
Ankle share of absorption (%)	23.2	24.1	—
Ankle +Work (J)	78.8	98.0	80.4%
Knee +Work (J)	82.5	98.3	84.0%

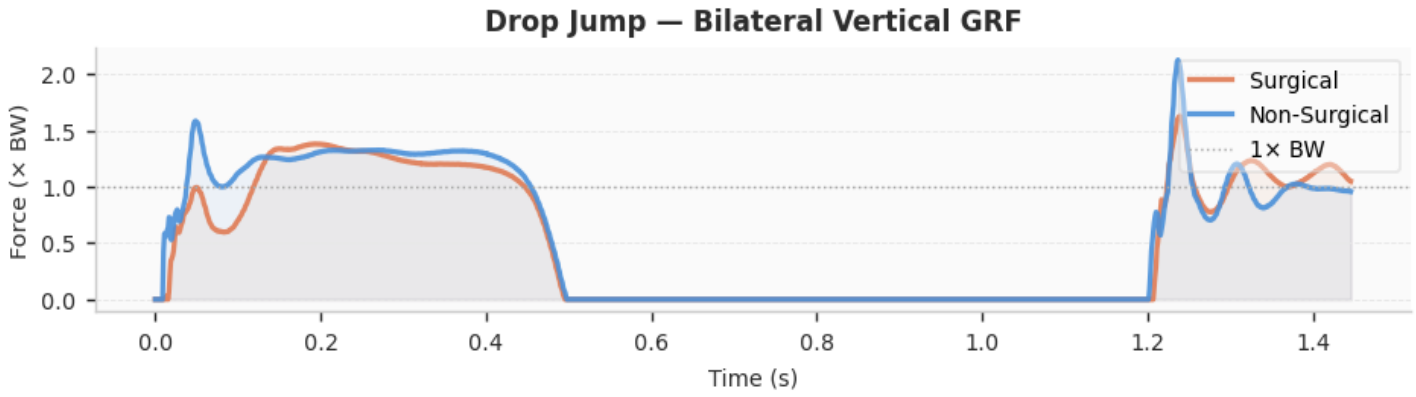
Hip +Work (J)	159.5	163.6	97.5%
Ankle share of +work (%)	24.4	27.1	—

CLINICAL INTERPRETATION

RSI 1.26 m/s with 0.48 s contact (⚠ below RTS target ≥ 1.30). Stiff-landing pattern likely — progressive plyometric loading indicated.

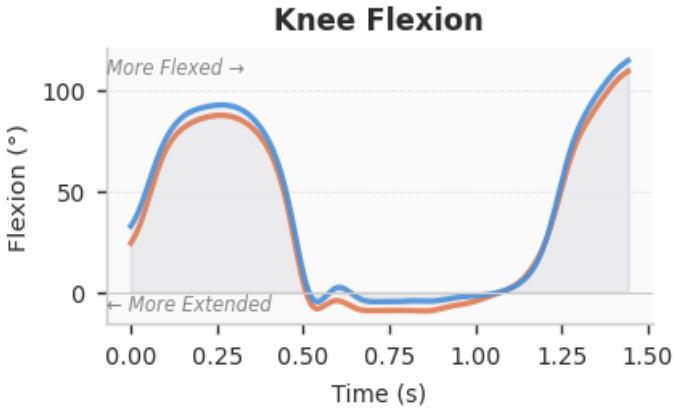
Landing symmetry 73% LSI: Surgical limb loading 27% less than non-surgical — surgical guarding detected.

Targets: RSI ≥ 1.30 m/s · Landing LSI 90–110% · Load rate LSI ≥ 90% · Knee flexion ≥ 60° · Valgus < 8° · Tibial IR < 10°.

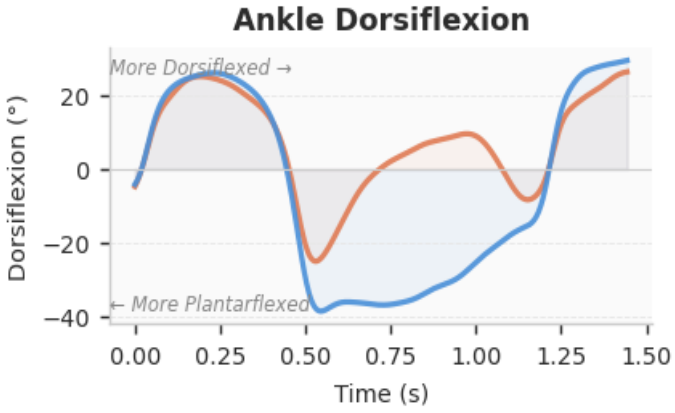


Bilateral GRF (normalised to BW). Orange = surgical, Blue = non-surgical.

Landing symmetry 73% at 200ms — surgical limb offloading detected.

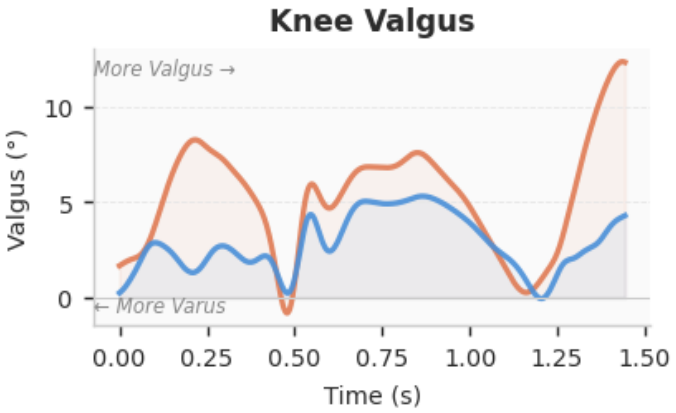


Knee flexion. Target ≥ 60° at contact. Reduced depth = stiff landing strategy.

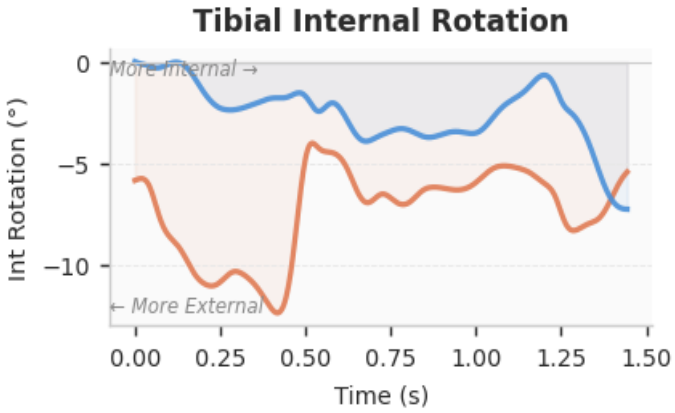


Ankle dorsiflexion. Restricted dorsiflexion → shifts load to quadriceps / increases ACL stress.

***Left panel:** Surgical limb knee flexion depth near-symmetric (LSI 94%) — approaching return-to-sport target; minor deficit remains.*



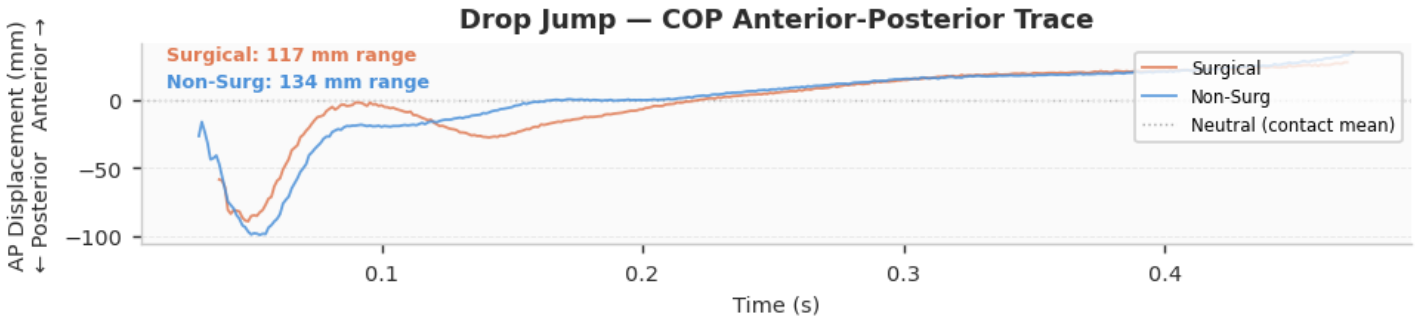
Knee valgus. Target < 8°. Elevated valgus = medial collapse / ACL re-injury risk.



Tibial internal rotation. IR > 10° at landing = increased ACL rotational load.

***Left panel:** Surgical knee valgus: 8.2° vs non-surgical 3.0° (Δ +5.2°) — mild side-to-side asymmetry; monitor closely.*

Right panel: *Surgical tibial IR: -6.4° vs non-surgical -0.4° (Δ -6.0°) — mild side-to-side asymmetry; monitor closely.*



AP COP displacement during the first contact (zero-centred). Smaller excursion on surgical limb = restricted fore-aft weight transfer / guarded landing strategy.

AP COP excursion: surgical 117 mm vs non-surgical 134 mm — less fore-aft travel on the surgical side — guarded or restricted loading.

Surgical limb bilateral landing symmetry demonstrates a significant deficit (LSI 73%) — targeted strengthening and re-testing indicated.

DROP LANDING

ABOUT THIS TEST

The Drop Landing isolates the deceleration phase of a jump: the athlete drops from a box and sticks the landing bilaterally without a subsequent jump. This removes the propulsive demand and focuses entirely on the athlete's ability to safely absorb and dissipate ground reaction forces. After ACL reconstruction, common compensations include a stiff-knee strategy (reduced flexion, sharp impact spike), excessive valgus collapse, and asymmetric weight acceptance—all of which increase mechanical stress at the reconstructed ligament. A smooth, progressive force curve with symmetric knee flexion ≥ 60° and valgus < 8° is the clinical target. Loading rate and impact transient LSI reflect side-to-side shock absorption capacity.

Metric	Surgical	Non-Surg	LSI / Δ
Peak GRF (N)	1339.1	1456.6	91.9%
Impulse (N·s)	621.7	640.8	97.0%
Loading Rate 50ms (N/s)	23242.3	22571.8	103.0%
Impact Transient (N)	708.4	1043.4	67.9%
Peak Knee Flexion (°)	111.1	117.3	94.8%
Peak Knee Valgus (°)	10.8	4.2	Δ +6.6°
Peak Tibial IR (°)	-4.6	1.0	Δ -5.5°
Tibial Rot Excursion (°)	5.4	4.3	Δ +1.1°
Peak Hip Flexion (°)	106.1	105.6	Δ +0.5°
Peak Hip Adduction (°)	-5.8	-7.8	Δ +2.0°
Peak Hip IR (°)	16.7	8.9	Δ +7.8°
Peak Ankle Dorsiflexion (°)	21.2	27.8	Δ -6.6°
Peak Plantarflexor Moment (Nm)	135.9	147.3	92.3%
Peak Knee Ext Moment (Nm)	102.0	137.2	74.3%
AP COP Excursion (mm)	73.3	106.4	—
Trials analysed	3	—	—
Ankle Absorption (J)	49.6	92.8	53.5%
Knee Absorption (J)	81.1	145.4	55.8%
Hip Absorption (J)	209.3	162.1	129.1%
Ankle share of absorption (%)	14.7	23.3	—

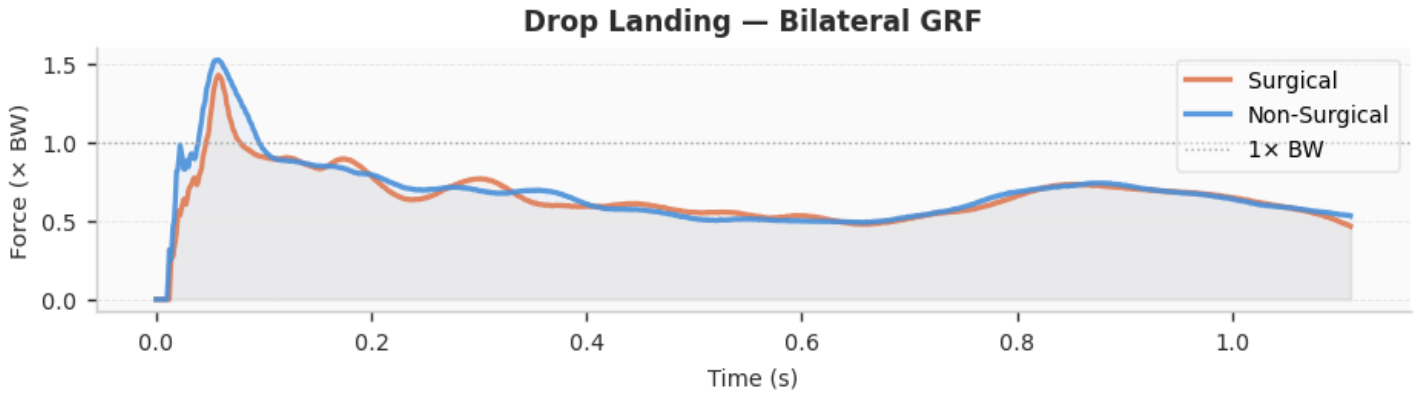
CLINICAL INTERPRETATION

Absorptive landing — assesses shock attenuation without a re-jump. Lower, more gradual force = better control.

Load rate LSI 103%: Symmetric loading strategy.

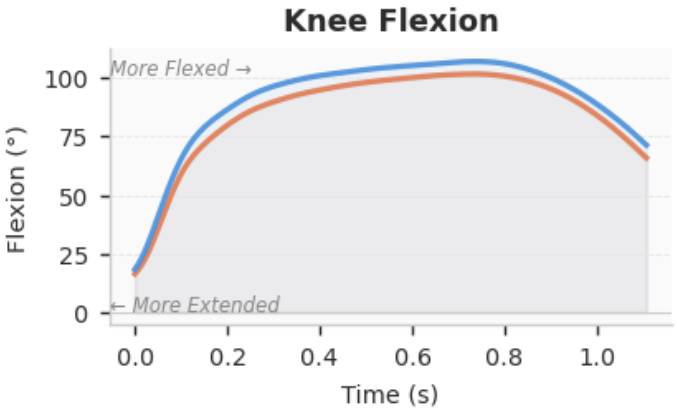
Impact transient LSI 68% — surgical limb initial impact 32% lower.

Targets: Peak GRF < 3× BW · Load rate LSI 90–110% · Impact transient LSI 90–110% · Knee flexion ≥ 60° · Valgus < 8° · Tibial IR < 10°.

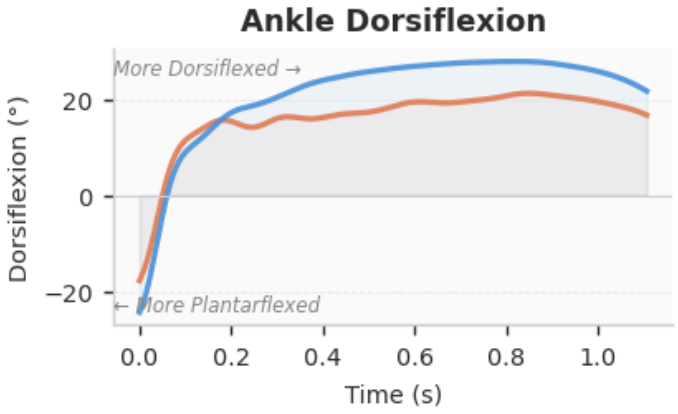


Bilateral GRF. Smooth progressive rise = good shock absorption. Sharp impact spike = stiff landing.

Peak GRF symmetry 92% — approaching return-to-sport target; minor deficit remains.

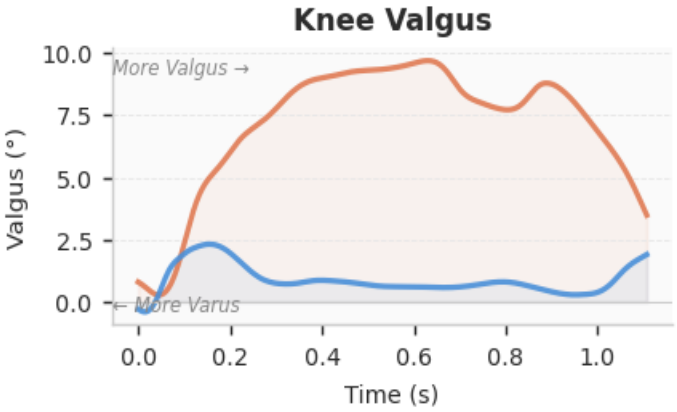


Knee flexion at landing. Reduced depth = stiff strategy / quad inhibition.

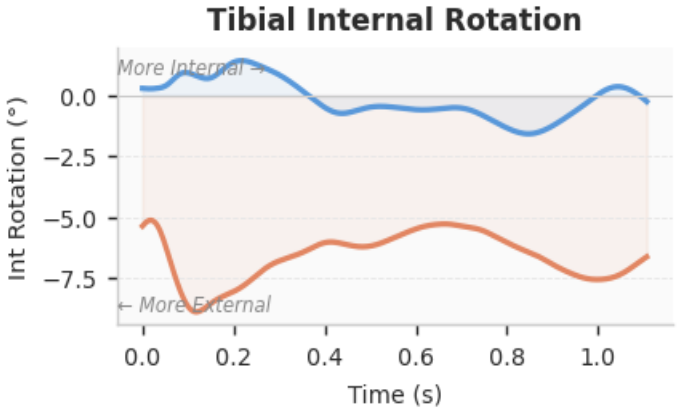


Ankle dorsiflexion. Restricted ROM limits knee flexion depth and increases ACL load.

***Left panel:** Surgical limb landing knee flexion depth near-symmetric (LSI 95%) — approaching return-to-sport target; minor deficit remains.*



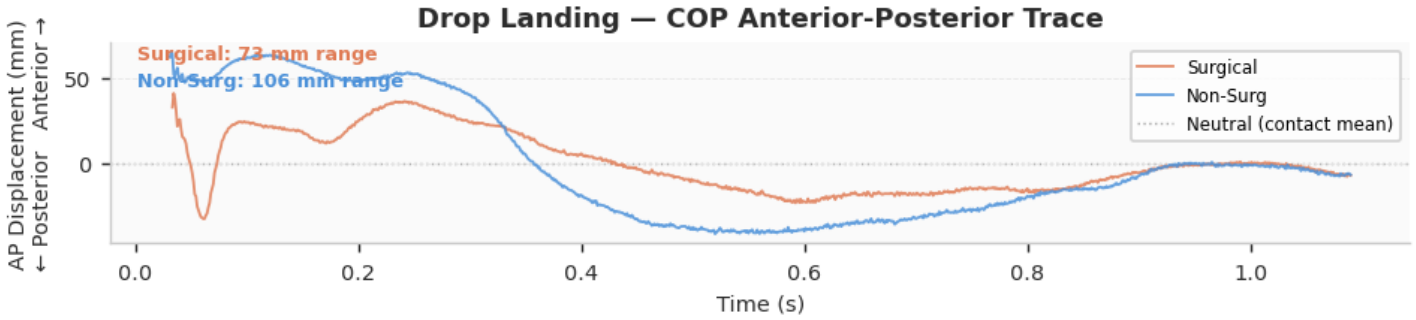
Knee valgus at landing. Target < 8°; higher surgical valgus = elevated ACL stress.



Tibial IR at landing. Combined with valgus = high ACL injury-risk pattern.

***Left panel:** Surgical knee valgus: 10.8° vs non-surgical 4.2° (Δ +6.6°) — side-to-side difference exceeds 6°; targeted intervention recommended.*

***Right panel:** Surgical tibial IR: -4.6° vs non-surgical 1.0° (Δ -5.5°) — mild side-to-side asymmetry; monitor closely.*



AP COP during landing contact (zero-centred). Posterior bias on surgical side = heel-dominant / reduced forefoot loading.

AP COP excursion: surgical 73 mm vs non-surgical 106 mm — less fore-aft travel on the surgical side — guarded or restricted loading.

Surgical limb peak landing force absorption near-symmetric (LSI 92%) — approaching return-to-sport target; minor deficit remains.

MAX VERTICAL JUMP (CMJ)

ABOUT THIS TEST

The Countermovement Jump (CMJ) is the gold-standard bilateral power test. The athlete begins in quiet stance, rapidly descends into a squat (countermovement), then drives upward for maximum jump height. The full GRF waveform reveals each phase: unweighting, eccentric loading, propulsion, and flight. After ACL reconstruction, typical deficits include an asymmetric countermovement depth (surgical limb offloading during the eccentric phase), reduced propulsive impulse, and side-to-side jump height asymmetry. Propulsive LSI < 90% is a common criterion for RTS decisions. Kinematic analysis identifies compensatory strategies—valgus, reduced knee flexion, and tibial rotation—that indicate the surgical limb is not being loaded with full confidence or mechanical efficiency.

Metric	Surgical	Non-Surg	LSI / Δ
Peak Propulsive Force (N)	1216.9	1296.9	93.8%
Net Propulsive Impulse (N·s)	147.9	189.1	78.2%
Peak Braking Force (N)	1217.9	1276.0	95.4%
Jump Height (in)	25.0	—	—
Flight Time (s)	0.719	—	—
Unweighting Impulse Surg/NS	82.7	79.2	—
Peak Knee Flexion (°)	84.1	88.0	95.6%
Peak Knee Valgus (°)	10.1	2.3	Δ +7.8°
Peak Tibial IR (°)	-4.8	2.7	Δ -7.5°
Tibial Rot Excursion (°)	7.5	6.1	Δ +1.4°
Peak Hip Flexion (°)	96.7	98.0	Δ -1.4°
Peak Hip Adduction (°)	-3.1	2.2	Δ -5.3°
Peak Hip IR (°)	11.4	1.8	Δ +9.6°
Peak Ankle Dorsiflexion (°)	28.0	29.6	Δ -1.6°
Peak Plantarflexor Moment (Nm)	163.2	181.4	90.0%
Peak Knee Ext Moment (Nm)	122.7	157.0	78.1%
AP COP Excursion (mm)	92.8	13.0	—
Trials analysed	2	—	—
Braking Impulse (N·s)	49.5	74.3	66.6%
Peak Power (W/kg)	78.7	—	—
Limb Peak Power (W)	3659.7	3915.2	93.5%

Limb Push-off Work (J)	559.9	636.5	88.0%
Ankle +Work (J)	85.0	114.8	74.1%
Knee +Work (J)	69.2	98.4	70.3%
Hip +Work (J)	265.7	233.6	113.7%
Ankle share of +work (%)	20.3	25.7	—

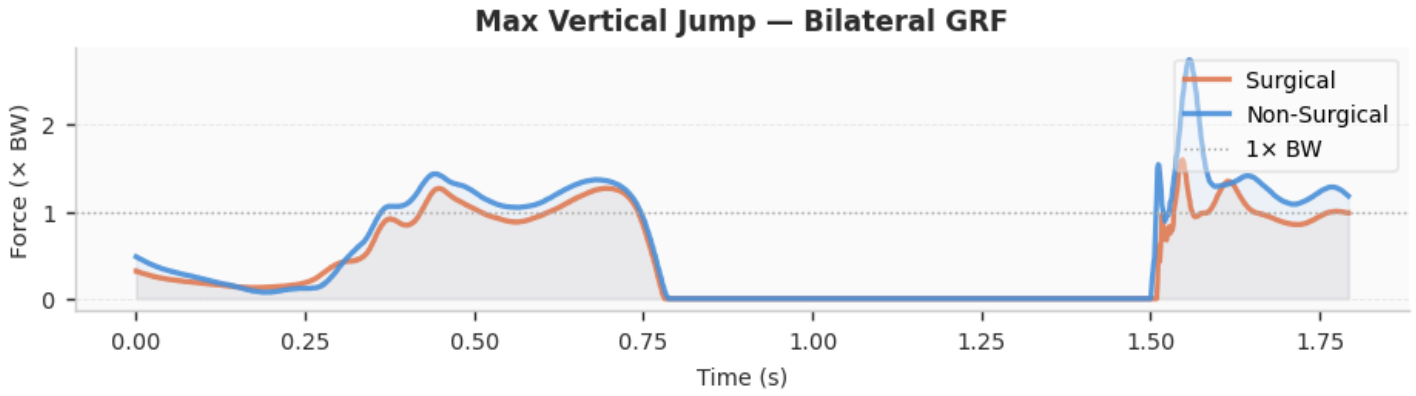
CLINICAL INTERPRETATION

Bilateral CMJ — sensitive to subtle propulsion deficits masked at lower intensities.

Braking force LSI 95%: Symmetric bilateral output.

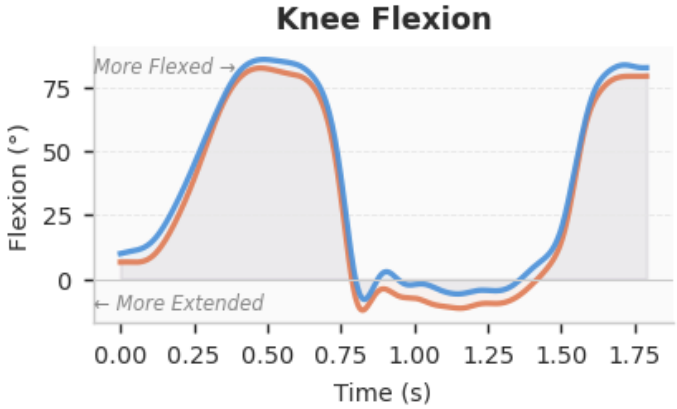
Propulsion LSI 78%. Targets: ≥ 90% propulsion, ≥ 90% peak force, matched unweighting impulse.

Knee +work LSI 70% vs hip 114%: The surgical knee contributes less push-off work while the hip makes up the difference — a hip-dominant strategy offloading the quadriceps.

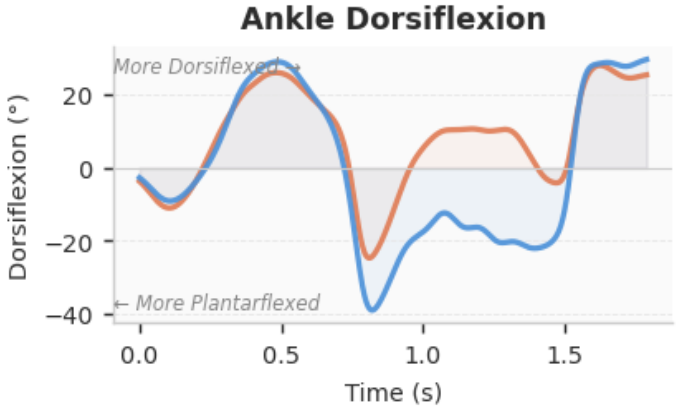


Full CMJ: unweighting → eccentric load → propulsion → flight. Propulsive peaks should be symmetric.

Surgical limb propulsive impulse shows a moderate asymmetry (LSI 78%) — continued loading progression recommended before clearance.

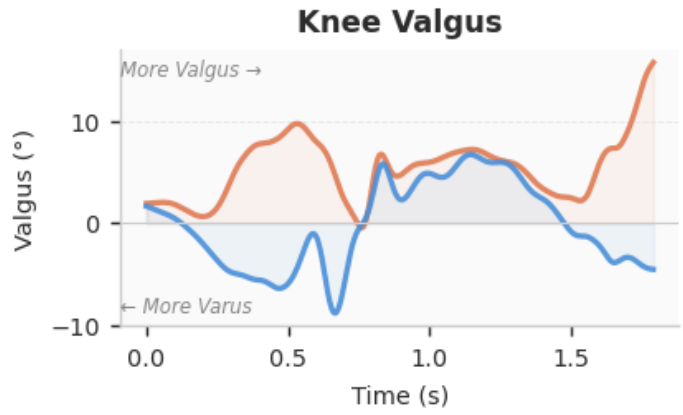


Knee flexion depth. Asymmetric countermovement depth = compensatory loading strategy.

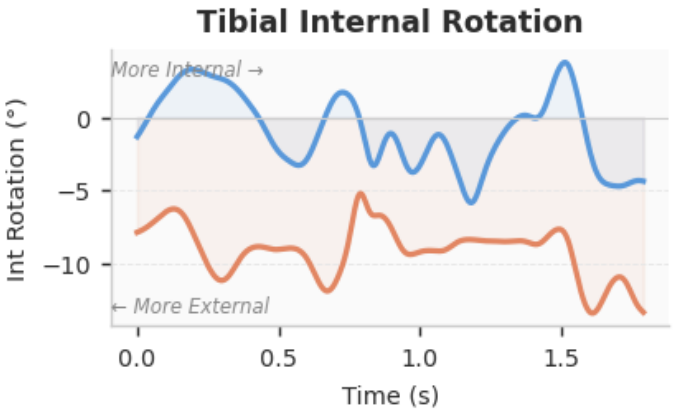


Ankle dorsiflexion. Restricted ROM reduces jump height and shifts load away from the knee.

***Left panel:** Surgical limb countermovement knee flexion depth is symmetric with the non-surgical side (LSI 96%) — meets return-to-sport threshold.*



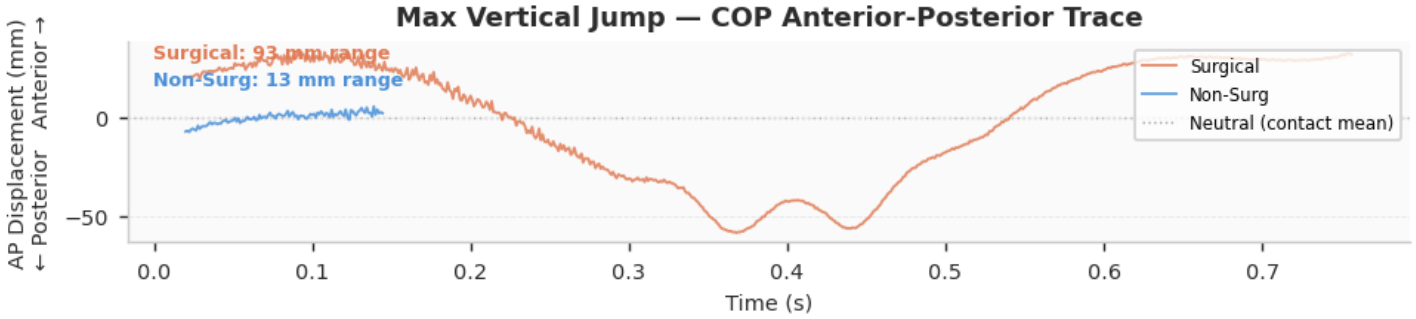
Knee valgus during propulsion. Dynamic valgus under load = ACL stress.



Tibial IR during propulsion. Coupled with valgus = high re-injury risk pattern.

Left panel: *Surgical knee valgus: 10.1° vs non-surgical 2.3° (Δ +7.8°) — side-to-side difference exceeds 6°; targeted intervention recommended.*

Right panel: *Surgical tibial IR: -4.8° vs non-surgical 2.7° (Δ -7.5°) — mild side-to-side asymmetry; monitor closely.*



AP COP during the propulsive contact (zero-centred). Anterior shift during propulsion drives jump height — asymmetric or reduced excursion on surgical side = limb offloading.

AP COP excursion: surgical 93 mm vs non-surgical 13 mm — more fore-aft travel on the surgical side — less controlled foot loading.

Surgical limb peak propulsive force is symmetric with the non-surgical side (LSI 95%) — meets return-to-sport threshold.

ENDURANCE SQUAT (30s)

ABOUT THIS TEST

The 30-Second Endurance Squat assesses neuromuscular fatigue and limb symmetry under sustained bilateral loading. The athlete performs continuous bilateral squats to a standardised depth for a full 30 seconds. Unlike single-repetition tests, this task reveals how well the neuromuscular system maintains symmetry as it fatigues. Athletes post-ACL reconstruction frequently show an initial LSI within acceptable range that deteriorates progressively across the set as the nervous system disinhibits protective guarding. The COP drift trace exposes this compensation: surgical-limb weight transfer decreases, traces diverge, and loading shifts toward the uninvolved side. This test is particularly sensitive to residual quadriceps inhibition that is not apparent during brief, high-energy tasks.

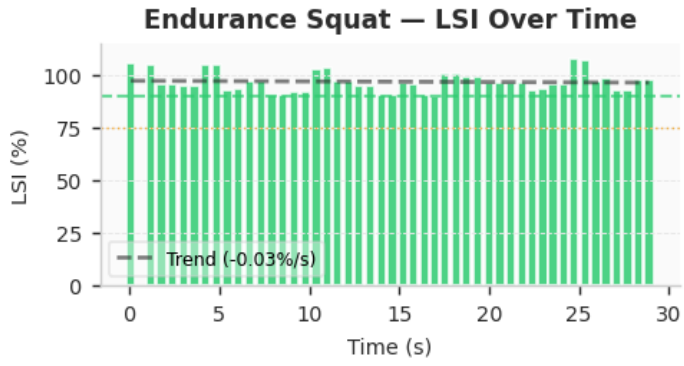
Metric	Surgical	Non-Surg	LSI / Δ
Mean Peak Force LSI (%) *(bilateral)*	—	—	97.1%
LSI First 10s (%) *(bilateral)*	—	—	97.0%
LSI Last 10s (%) *(bilateral)*	—	—	97.4%
Fatigue Drift (Δ%)	0.4	—	—
Total Reps Detected	48.0	—	—
Mean Peak Force Surg (N)	903.3	—	—
Mean Peak Force NS (N)	931.9	—	—
AP COP Excursion — Surg / NS (mm)	218.2	225.8	—
Peak Knee Ext Moment (Nm)	123.6	143.1	86.4%

CLINICAL INTERPRETATION

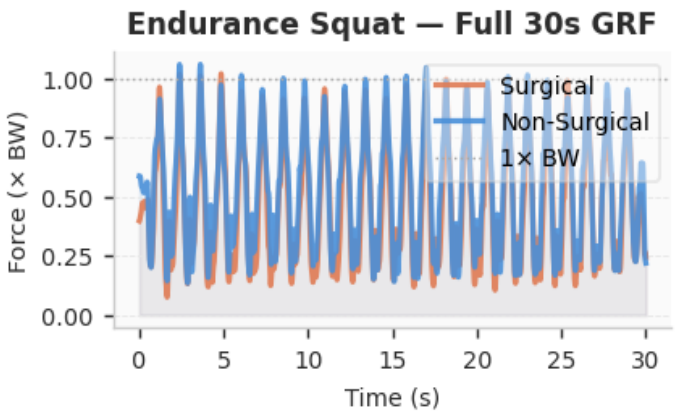
30s bilateral squat — reveals neuromuscular fatigue-induced asymmetry missed by single-rep testing.

Fatigue drift 0.4%: ✓ Symmetry maintained under fatigue.

Targets: Mean LSI ≥ 90% · Fatigue drift < 5% · No progressive offloading across 30s.

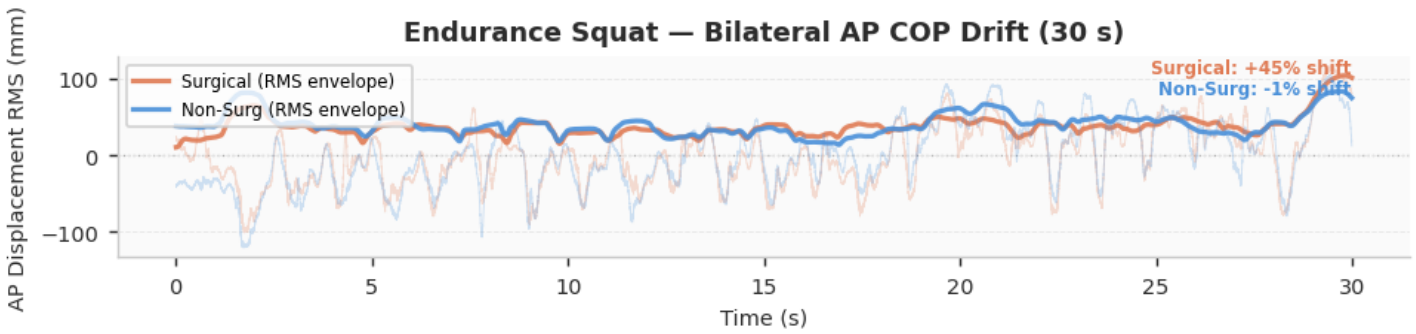


Per-rep LSI across 30s. Declining trend = neuromuscular fatigue breakdown.



Full 30s GRF trace. Force magnitude and rhythm across the set.

***Left panel:** Surgical limb mean 30s force symmetry is symmetric with the non-surgical side (LSI 97%) — meets return-to-sport threshold.*



Bilateral AP COP drift across 30s (zero-centred, 1s RMS envelope). Surgical-side amplitude shrinking or trending posterior = offloading under fatigue. Diverging traces = progressive asymmetry with sustained effort.

Fatigue drift: 0.4% LSI change across 30s — symmetry maintained under sustained effort.

Surgical limb mean force symmetry over 30 s is symmetric with the non-surgical side (LSI 97%) — meets return-to-sport threshold.

SINGLE-LEG VERTICAL JUMP

ABOUT THIS TEST

The Single-Leg Vertical Jump isolates each limb independently, removing the ability of the uninvolved leg to compensate. The athlete performs a maximal countermovement jump landing on the same leg. Jump height LSI (surgical vs. non-surgical) is among the most sensitive power-based RTS criteria—a threshold of ≥ 90% is widely cited in ACL literature. Single-leg testing also reveals subtle compensations masked in bilateral tasks: reduced propulsive depth, hip drop (Trendelenburg pattern), and increased tibial rotation are common markers of residual neuromuscular asymmetry.

Metric	Surgical	Non-Surg	LSI / Δ
Jump Height (in)	11.9	13.9	85.7%
Peak GRF (N)	2026.3	1869.0	108.4%
Net Impulse (N·s)	233.6	252.3	92.6%
Peak RFD (N/s)	9492.0	10472.1	—
Contact Time (s)	0.96	0.89	—
Peak Knee Flexion (°)	54.0	69.1	—
Peak Knee Valgus (°)	2.5	7.3	Δ -4.7°
Peak Tibial IR (°)	-2.3	3.1	Δ -5.4°
Peak Ankle Dorsiflexion (°)	22.0	30.9	Δ -8.8°
Peak Plantarflexor Moment (Nm)	264.5	236.3	111.9%
Peak Hip Adduction (°)	8.2	11.1	Δ -3.0°
Peak Knee Ext Moment (Nm)	161.5	231.0	69.9%
AP COP Excursion (mm)	94.9	154.6	—
Trials analysed	3	3	—
Peak Power (W/kg)	45.2	47.7	94.8%
Push-off Work (J)	769.5	978.4	78.6%
Peak RPD (W/s)	28221.2	23610.2	119.5%
Ankle +Work (J)	90.0	161.7	55.7%
Knee +Work (J)	44.1	74.6	59.1%
Hip +Work (J)	301.5	332.2	90.8%
Ankle share of +work (%)	20.7	28.5	—

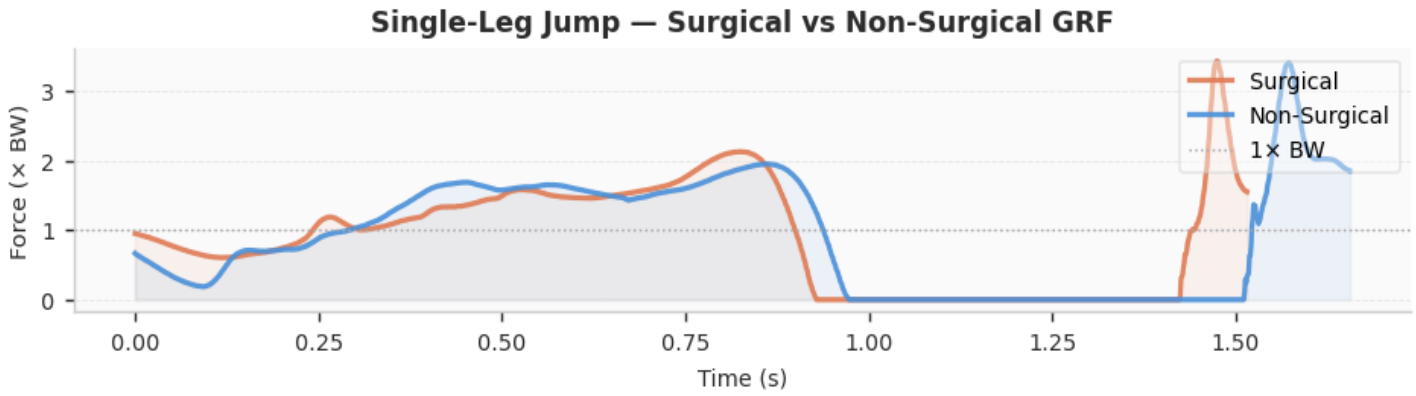
CLINICAL INTERPRETATION

Highest-demand test — isolates each limb, removing bilateral compensation. Strongest re-injury risk predictor.

Jump height LSI 86%: ⚠ 14% below non-surgical. Continued plyometric loading required.

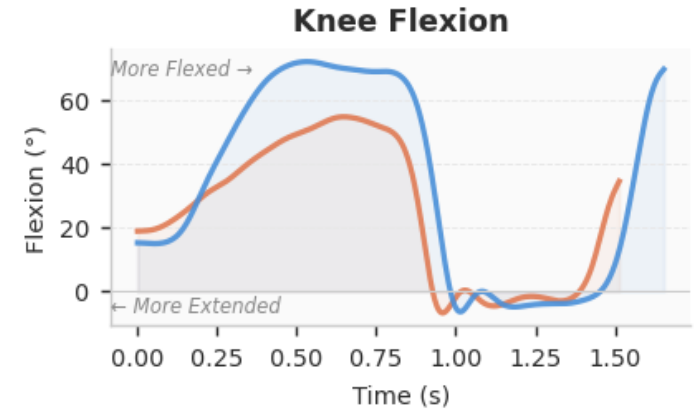
Knee +work LSI 59% vs hip 91%: The surgical knee does less push-off work while the hip makes up the difference — quadriceps avoidance under single-leg load.

Targets: Jump height LSI ≥ 90% · Peak GRF LSI ≥ 90% · Valgus < 8° · Tibial IR < 10° on surgical limb.

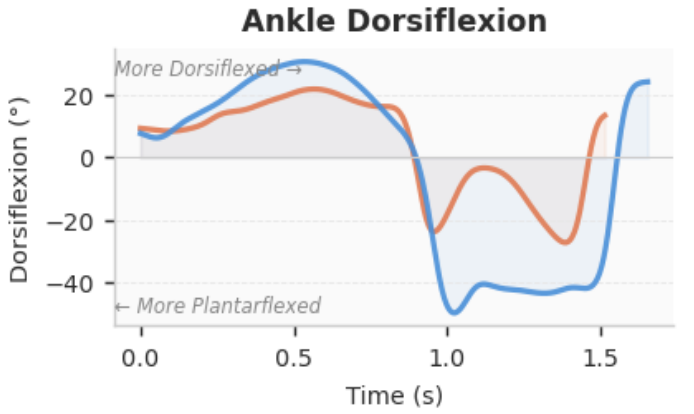


Per-limb GRF (separate trials). Propulsive peak height and shape should match.

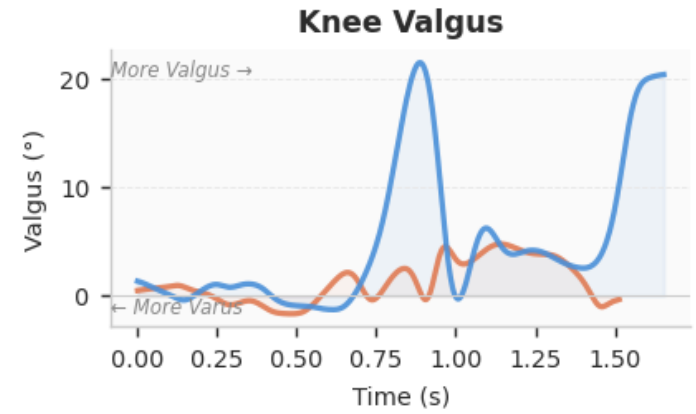
Surgical limb single-leg jump height shows a moderate asymmetry (LSI 86%) — continued loading progression recommended before clearance.



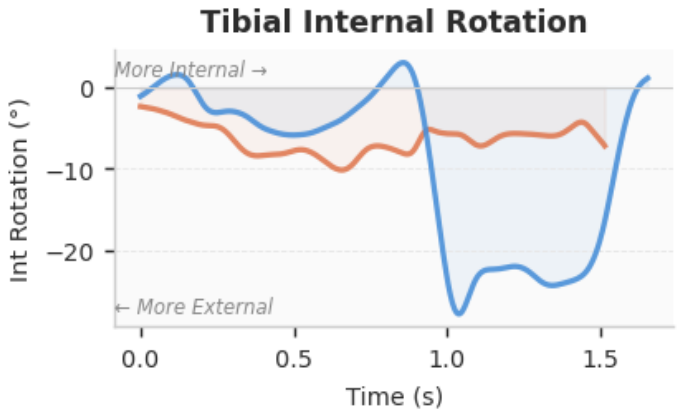
Knee flexion per limb. Reduced surgical depth = power deficit or guarded strategy.



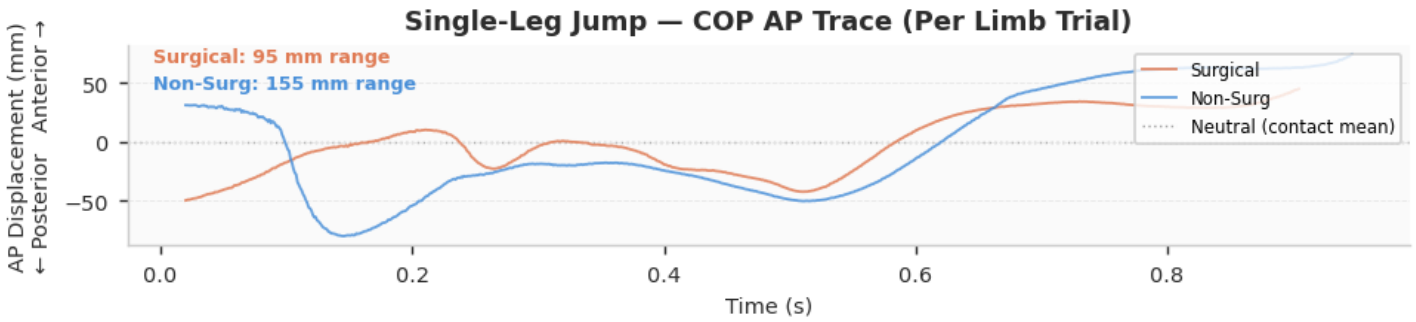
Ankle dorsiflexion per limb. Restricted ROM = reduced propulsive range / increased ACL load.



Knee valgus per limb. Elevated surgical valgus under single-leg load = re-injury risk.



Tibial IR per limb. Surgical-side IR > 10° = high rotational ACL stress pattern.



AP COP during the take-off contact (zero-centred, per limb trial). Reduced anterior excursion on surgical side = guarded takeoff / limited push-off.

AP COP excursion: surgical 95 mm vs non-surgical 155 mm — less fore-aft travel on the surgical side — guarded or restricted loading.

Surgical limb single-leg jump height shows a moderate asymmetry (LSI 86%) — continued loading progression recommended before clearance.

BALANCE — STANDARD (EYES CLOSED, FIRM SURFACE)

ABOUT THIS TEST

The Balance / Proprioception test assesses joint position sense and neuromuscular control under sensory challenge. The athlete stands on a single leg with eyes closed, removing visual compensation and isolating the mechanoreceptor feedback system of the ankle and knee. Center of Pressure (COP) velocity is the most sensitive metric: a higher velocity indicates greater corrective activity, meaning the neuromuscular system is working harder to maintain stability. After ACL reconstruction, deficits in mechanoreceptor density within the graft and capsular tissue result in elevated COP velocity and a larger sway ellipse on the surgical limb. The Airex pad condition adds an unstable surface to further stress the proprioceptive system, revealing deficits that may be masked on firm ground. Symmetry between limbs (LSI ≥ 85–90%) is the primary RTS criterion for this test.

Metric	Surgical	Non-Surg	LSI / Δ
Stance hold	FAIL	FAIL	—
COP Mean Velocity (mm/s)	380.0	140.5	**37.0%**
COP Ellipse Area (mm²)	40703.1	3873.8	**9.5%**
Total Excursion (mm)	2712.6	614.8	—
AP Range (mm)	168.6	106.0	—
ML Range (mm)	366.4	45.0	—
RMS Displacement (mm)	74.7	22.0	—

CLINICAL INTERPRETATION

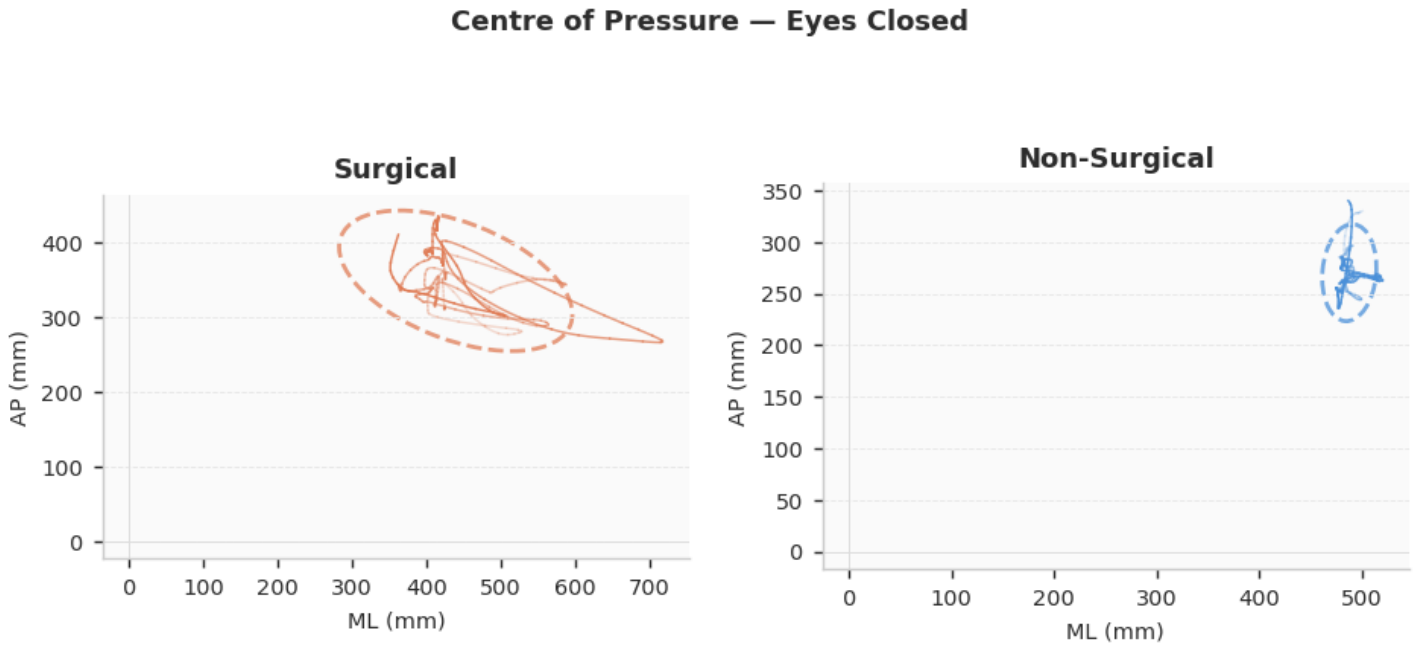
Eyes-closed single-leg stance on firm surface. COP velocity is the most sensitive metric — higher speed = more corrective activity = reduced proprioception.

Surgical 380 mm/s vs Non-Surgical 140 mm/s — Surgical limb less stable. Proprioceptive retraining indicated.

COP Velocity LSI 37% (NS÷Surg×100; below 100% = surgical side sways more). LSI target ≥ 90%. COP low-pass filtered at 10 Hz.

✗ **Failed stance hold:** surgical limb — foot moved + touchdown; non-surgical limb — touchdown. The athlete could not hold single-leg eyes-closed stance without moving the foot or touching down; this counts as a fail (sway values below include the loss of balance).

Surgical limb failed: velocity symmetry is capped at 50% for scoring — the non-surgical limb also failed, so this is a bilateral balance deficit, not just an asymmetry.



COP trajectory (10 Hz low-pass) — eyes closed. Tighter ellipse = steadier stance. 95% confidence ellipse shown.

BALANCE — AIREX PAD (EYES CLOSED, UNSTABLE SURFACE)

ABOUT THIS TEST

The Balance / Proprioception test assesses joint position sense and neuromuscular control under sensory challenge. The athlete stands on a single leg with eyes closed, removing visual compensation and isolating the mechanoreceptor feedback system of the ankle and knee. Center of Pressure (COP) velocity is the most sensitive metric: a higher velocity indicates greater corrective activity, meaning the neuromuscular system is working harder to maintain stability. After ACL reconstruction, deficits in mechanoreceptor density within the graft and capsular tissue result in elevated COP velocity and a larger sway ellipse on the surgical limb. The Airex pad condition adds an unstable surface to further stress the proprioceptive system, revealing deficits that may be masked on firm ground. Symmetry between limbs (LSI ≥ 85–90%) is the primary RTS criterion for this test.

Metric	Surgical	Non-Surg	LSI / Δ
Stance hold	FAIL	FAIL	—
COP Mean Velocity (mm/s)	315.0	225.1	50.0%
COP Ellipse Area (mm²)	33657.9	17728.4	52.7%
Total Excursion (mm)	3634.9	2893.3	—
AP Range (mm)	195.1	191.3	—
ML Range (mm)	221.2	137.2	—
RMS Displacement (mm)	62.8	44.4	—

CLINICAL INTERPRETATION

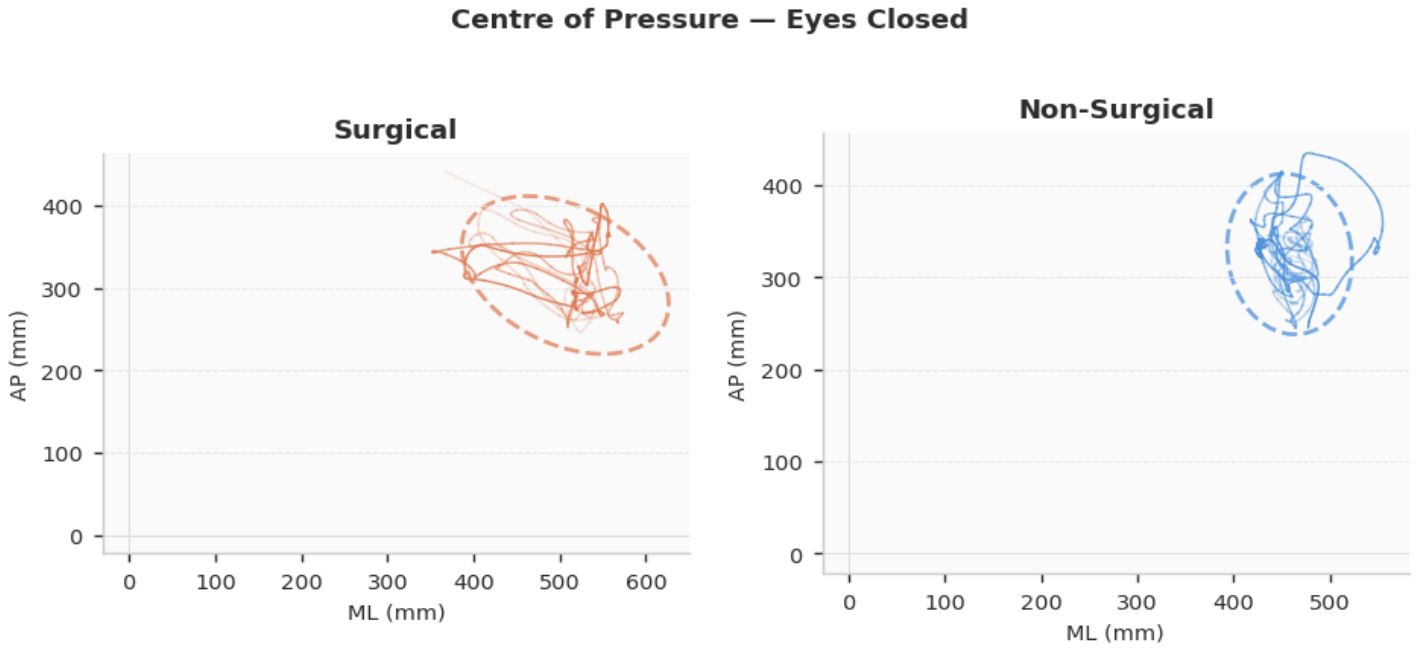
Eyes-closed single-leg stance on Airex foam pad. Higher COP values are expected — the unstable surface challenges proprioceptive and neuromuscular systems. Symmetry between limbs (LSI ≥ 85%) is the key target.

Surgical 315 mm/s vs Non-Surgical 225 mm/s — Surgical limb less stable. Proprioceptive retraining indicated.

COP Velocity LSI 50% (NS÷Surg×100; below 100% = surgical side sways more). LSI target ≥ 90%. COP low-pass filtered at 10 Hz.

✗ **Failed stance hold:** surgical limb — foot moved; non-surgical limb — foot moved. The athlete could not hold single-leg eyes-closed stance without moving the foot or touching down; this counts as a fail (sway values below include the loss of balance).

Surgical limb failed: velocity symmetry is capped at 50% for scoring — the non-surgical limb also failed, so this is a bilateral balance deficit, not just an asymmetry.



COP trajectory (10 Hz low-pass) — eyes closed. Tighter ellipse = steadier stance. 95% confidence ellipse shown.